

Inventory Management System

Inventory Operations & Replenishment Policy

1. Purpose and Scope

This policy defines the standard operational rules and guidelines for managing products, stock levels, purchases, sales, suppliers, and inventory replenishment within the Inventory Management System. It serves as a definitive knowledge base for internal system operations, providing structured business rules designed to be retrieved and utilized by an integrated Artificial Intelligence (AI) assistant to accurately answer internal employee queries.

2. User Roles

The system is restricted to internal operations only and is not accessible by external customers. Access and permissions are defined by the following roles:

Role	Responsibilities
Cashier	Responsible for recording Sales transactions and checking Product availability during checkout processes.
Inventory Staff	Responsible for monitoring physical stock, logging Purchases, and proactively tracking low-stock Products to ensure uninterrupted operations.
Manager	Oversees comprehensive inventory operations; monitors aggregate Sales, Purchases, Suppliers, overall stock levels, and system analytics.

3. Product and Stock Rules

The foundational elements of the system revolve around **Products** and their classifications (**Categories**). The following rules apply:

- **Categorization:** Every Product belongs to a Category.
- **Product Attributes:** Each Product must possess a unique SKU (Stock Keeping Unit), a name, a unit price, a stock quantity, and a defined low-stock threshold.
- **Stock Definition:** The "stock quantity" represents the exact number of units currently available in inventory.
- **Non-Negative Constraint:** Stock quantity must never become negative under any circumstances.

- **Low-Stock Status:** A Product is classified as "low-stock" when its stock quantity is less than or equal to its low-stock threshold.
- **Out-of-Stock Status:** A Product is classified as "out of stock" when its stock quantity reaches exactly zero.

4. Sales Rules

Transactions initiated by customers (but recorded by internal staff) are processed through **Sales**. The processing rules are as follows:

- **Composition:** Every Sale consists of one or more **Sales Items**.
- **Sales Item Data:** Each Sales Item references a specific Product and specifies the quantity sold. The selling price stored in the Sales Item represents the exact price of the Product at the time of the transaction.
- **Stock Validation:** A Sale cannot be completed if the requested quantity for any item is greater than the available stock for that Product.
- **Stock Deduction:** When a Sale is successfully completed, the sold quantity must be immediately deducted from the respective Product's stock.
- **Condition of Deduction:** Stock deduction must only occur for successfully completed Sales; pending or failed transactions do not alter stock quantities.

5. Purchase Rules

Stock acquisitions are managed through **Purchases**. The following rules govern inbound inventory:

- **Supplier Association:** Every Purchase must be explicitly associated with a **Supplier**.
- **Composition:** Every Purchase contains one or more **Purchase Items**.
- **Purchase Item Data:** Each Purchase Item references a specific Product, specifying the quantity acquired and the unit cost.
- **Total Calculation:** The total cost of a Purchase is calculated as the sum of (Quantity × Unit Cost) for all associated Purchase Items.
- **Stock Addition:** When a Purchase is successfully completed and recorded, the purchased quantity must be immediately added to the respective Product's stock.

6. Low-Stock and Replenishment Rules

To ensure business continuity, replenishment protocols must be strictly followed:

- **Identification:** Products with a Stock Quantity $\leq$ Low-Stock Threshold must be immediately reviewed for replenishment.
- **Priority Hierarchy:** Products with zero stock (Out of Stock) receive the absolute highest replenishment priority.

- **Action Requirement:** Products at or below their low-stock threshold should be actively considered for a new Purchase.
- **Decision Factors:** Replenishment decisions should factor in current stock levels and recent Sales activity, where such historical data is available.

Important AI Distinction: The AI assistant must distinctly differentiate between "low stock" (quantity $\leq$ threshold, but > 0) and "out of stock" (quantity = 0).

7. Supplier Rules

Suppliers represent the external vendors providing inventory to the system:

- **Data Retention:** Each Supplier record stores the vendor's name and contact information.
- **Transaction History:** A single Supplier can have multiple Purchase transactions associated with them.
- **Sourcing Reference:** Employees use Supplier information to determine where a Product can be sourced.
- **Product Linkage:** Employees must be able to seamlessly identify the Suppliers associated with the Products they provide based on historical Purchases.

8. Dashboard and Analytics Rules

The system provides operational metrics to assist management and staff in decision-making. The following metrics are standard:

Metric	Business Definition
Total Products	The total number of unique Products managed within the system.
Total Categories	The total number of product classifications available.
Total Suppliers	The total number of registered vendors in the system.
Total Stock Quantity	The aggregate sum of all available units across all Products.
Low-Stock Products	The count or list of Products currently at or below their respective low-stock thresholds.
Total Purchases	The aggregate number of completed inbound Purchase transactions.
Total Sales	The aggregate number of completed outbound Sales transactions.
Recent Activity	A chronological log of the latest Sales and Purchases.

Metric	Business Definition
Most Sold Products	A ranking of Products based on the highest aggregate quantities sold across all Sales Items.

9. AI Assistant Usage Guidelines

The integrated AI assistant is designed to answer operational questions. Example queries include:

- "Which products are low in stock?"
- "Which products are out of stock?"
- "What products need replenishment?"
- "What are the top-selling products?"
- "Who supplies this product?"
- "How much stock do we currently have?"
- "What were the recent purchases?"
- "What were the recent sales?"
- "What is the rule for low-stock products?"

10. Important AI Behavior

The AI assistant must strictly adhere to the following behavioral guidelines to ensure data integrity and user trust:

- **No Hallucination:** The AI assistant must never invent inventory numbers, product quantities, suppliers, sales, or purchases.
- **Live Data Access:** The AI must use live system data or API results for any questions regarding current inventory levels and actual transactions.
- **Policy Adherence:** The AI must use this document strictly as a knowledge source for business rules and operational policies.
- **Source Distinction:** The AI must clearly distinguish between static policy information (retrieved from this document) and real-time operational information (retrieved from the live system).
- **Graceful Failure:** If the required information is unavailable in the database or policy document, the AI must explicitly state that it cannot determine the answer from the available data.